WEEK 2 OF 15 · STAGE 1: FOUNDATIONS

Climate primer

How CO2 becomes warming, and why the total matters

New R skill Vectors and data frames; reading CSV files; a first ggplot2 chart

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 — Foundations

- **Week 1** — Why model the future?
- **Week 2** — **Climate primer**
- **Week 3** — Energy primer
- **Week 4** — Economics primer
- **Week 5** — Cost-benefit IAMs: DICE
- **Week 6** — The scenario framework

Stage 2 — GCAM in depth

- **Week 7** — Inside GCAM
- **Week 8** — Energy in GCAM
- **Week 9** — Land, water and bioenergy
- **Week 10** — Emissions and policy

Stage 3 — Advanced methods

- **Week 11** — Optimisation IAMs
- **Week 12** — Why models disagree
- **Week 13** — Uncertainty
- **Week 14** — Frontiers and limits
- **Week 15** — Projects and review

TODAY'S GOALS

By the end of today you can

1. Trace the four steps from emissions to temperature
2. Explain why warming follows the running total of CO2
3. Work out how long a carbon budget lasts
4. Read a CSV file into R and draw a first chart with ggplot2

? **Big question** Does a tonne of CO2 burned in 1960 still matter today?

Does old CO2 still count?

A tonne of CO2 burned in 1960 still warms the planet today about as much as a tonne burned this morning.



If that is true, what kind of limit does a temperature target set on our emissions?

AN EVERYDAY PICTURE

The atmosphere is a bathtub



The tap is the CO2 we emit each year. The plug hole is small: oceans and forests take up some CO2, but a little under half of each year's emissions stays in the air. The water level keeps rising as long as the tap runs faster than the drain.

What it means for models

1. Tap = yearly emissions
2. Water level = CO2 in the air
3. Only turning the tap (almost) off stops the rise

HOW IT WORKS

From emissions to temperature in four steps

1

1 Emissions

Fossil fuels and land clearing release CO2 each year

2

2 Concentration

A little under half stays in the air: 280 to over 420 ppm

3

3 Forcing

Extra CO2 traps heat, measured in W/m²

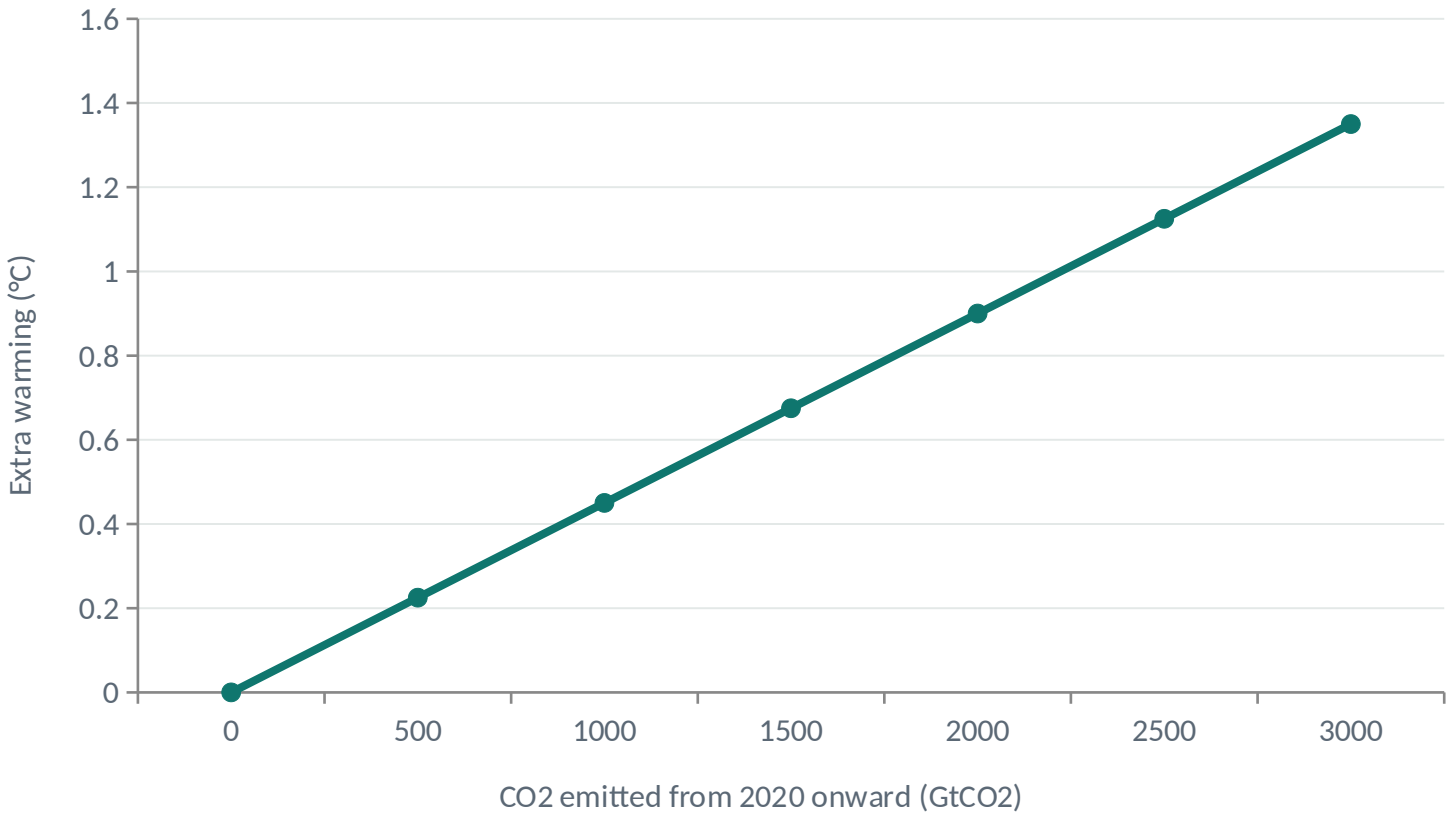
4

4 Temperature

The surface warms until energy in and out balance

Steps 2 to 4 are what the Hector model calculates. GCAM hands its emissions to Hector in week 10.

Warming tracks the running total of CO2



Computed from ΔT ≈ 0.45 °C × cumulative CO2 ÷ 1,000 Gt

Read it like this

1. Each 1,000 GtCO2 adds about 0.45 °C
2. A straight line: no 'safe' emissions that do not count
3. Warming stops rising only at net zero

NUMBERS TO REMEMBER

Two carbon budgets to remember

500 Gt

CO_2 from 2020 for a 50% chance of 1.5 °C

1,150 Gt

CO_2 from 2020 for a 67% chance of 2 °C

~40 Gt

CO_2 the world emits each year

Budgets from IPCC AR6 Working Group I (2021), counted from the start of 2020; part is already used.

Worked example: how long does 500 Gt last?

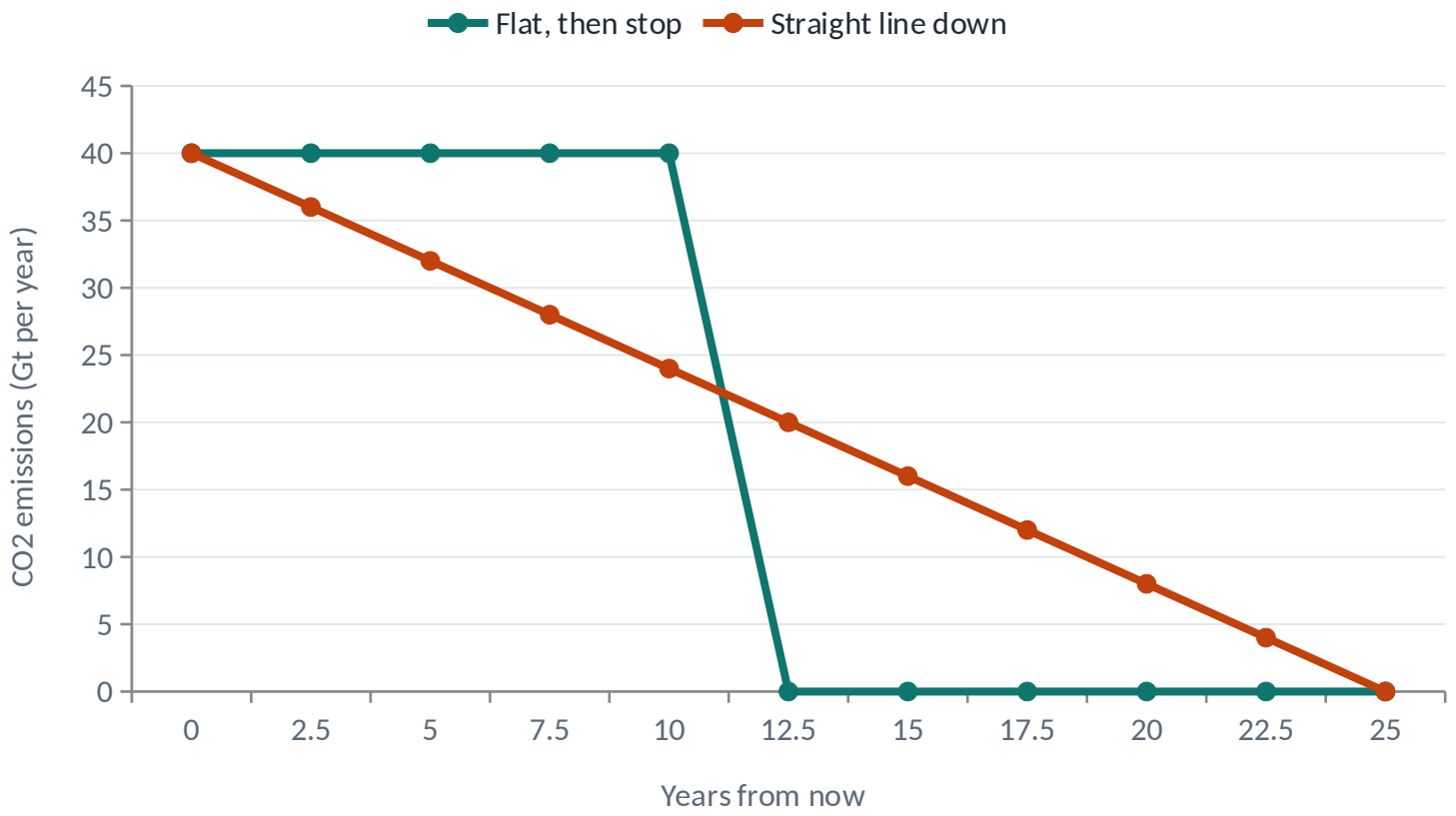
Problem The 1.5 °C budget is 500 GtCO2. The world emits about 40 Gt a year. How long does the budget last?

1. If emissions stay flat: 500 ÷ 40 = 12.5 years, then a sudden stop
2. If they fall in a straight line to zero over T years, the total is a triangle
3. Triangle area = ½ × 40 × T
4. Set it equal to the budget: ½ × 40 × T = 500, so 20T = 500
5. T = 500 ÷ 20 = 25 years

Answer **12.5 years if flat; 25 years with a steady straight-line decline**

PICTURE THE NUMBERS

Same budget, two shapes



Illustrative paths for the 500 Gt budget at 40 Gt a year

Read it like this

1. The area under each line is the CO2 emitted
2. Both areas equal 500 Gt
3. Both end at zero

Quick check

The 2 °C budget is 1,150 GtCO2. At 40 Gt a year, how long does it last if emissions stay flat?

- **A** About 12 years
- **B** About 29 years
- **C** About 58 years
- **D** About 115 years

Quick check: answer

The 2 °C budget is 1,150 GtCO2. At 40 Gt a year, how long does it last if emissions stay flat?

A About 12 years

B **About 29 years** ✓

C About 58 years

D About 115 years

COMPARE

Two numbers people mix up

Emissions

- What we add each year
- Measured in Gt of CO2 per year
- About 40 Gt per year today
- The tap in the bathtub

Concentration

- What is in the air now
- Measured in parts per million (ppm)
- About 280 before industry, over 420 today
- The water level in the bathtub

WATCH OUT

Common mix-ups

Not quite	Better
Halve emissions and you halve warming.	Halving the yearly rate only slows the rise. Warming stops rising only at net zero.
ppm and Gt per year measure the same thing.	ppm is what is in the air (the level). Gt per year is what we add (the tap).
500 Gt of CO2 gives 1.5 °C of warming on its own.	It gives about 0.225 °C more. We are already near 1.1 °C, and room is kept for methane and other gases.

Your first chart in R

- A data frame is a table; each column is a vector
- `$` picks one column out of a table
- Every ggplot: data, aes (what goes on which axis), a geom

```
co2 <- read_csv("data/co2_annual.csv")
glimpse(co2)
max(co2$ppm)

ggplot(co2, aes(x = year, y = ppm)) +
  geom_line() +
  labs(y = "CO2 (ppm)")
```

Words to add to the class glossary

Concentration How much CO2 is in the air, measured in parts per million (ppm). About 280 before industry, over 420 today.

Radiative forcing Extra heat trapped by greenhouse gases, measured in watts per square metre (W/m²). A thicker 'blanket'.

Cumulative emissions All the CO2 emitted added up over the years: the running total.

Carbon budget The total CO2 that can still be emitted while keeping warming below a chosen limit.

Net zero The point where CO2 added to the air each year is balanced by CO2 removed, so the running total stops growing.

HANDS-ON LAB

Lab 2: first plots and the Hector climate model

Read a CO2 data file and plot it, then run the Hector climate model for a low and a high emissions path and compare the temperatures.

Script: labs/lab2.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

LAB · STEP 1 OF 4

Read and look at the data

read_csv() loads a table. glimpse() and head() show what is inside before you plot anything.

```
library(tidyverse)

co2 <- read_csv("data/co2_annual.csv")
glimpse(co2)
head(co2)
max(co2$ppm)
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Draw the CO2 chart

Map year to x and ppm to y, then draw a line.

```
ggplot(co2, aes(x = year, y = ppm)) +
  geom_line() +
  labs(y = "CO2 (ppm)")
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Run Hector for a low path

Hector is the simple climate model inside GCAM. newcore() sets it up, run() runs it, fetchvars() pulls out temperature.

```
library(hector)
ini <- system.file("input/hector_ssp126.ini",
                   package = "hector")
low <- newcore(ini, name = "low")
run(low)
t_low <- fetchvars(low, 2000:2100,
                   GLOBAL_TAS())
```

Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.

Run a high path and compare

Repeat with SSP5-8.5, stack the two results and colour the lines by scenario.

```
ini <- system.file("input/hector_ssp585.ini",
                   package = "hector")
high <- newcore(ini, name = "high")
run(high)
t_high <- fetchvars(high, 2000:2100,
                    GLOBAL_TAS())
temps <- bind_rows(t_low, t_high)
ggplot(temps, aes(year, value,
                  colour = scenario)) +
  geom_line()
```

Did it work?

You should see

- Your CO2 chart rises over time and ends above 400 ppm
- The two Hector lines are close until about 2040, then split
- By 2100 the high path is several degrees warmer than the low path

Worksheet questions

1. Why do the two temperature paths differ so little by 2050? Use the idea of cumulative emissions.
2. What is the highest CO2 value in the file, and in which year?
3. Change geom_line() to geom_point(). What changed and what stayed the same?

RECAP

Take-home messages

1. Emissions raise CO2 in the air, which traps heat (forcing), which warms the surface
2. Warming follows the running total of CO2: about 0.45 °C per 1,000 Gt
3. So a temperature limit becomes a carbon budget, and warming stops only at net zero
4. In R: read_csv() loads a table, ggplot() + aes() + a geom draws it

Next: week 3

Week 3 asks where the emissions come from: energy, and the four factors of the Kaya identity.

Practise

Interactive course, week 2: try the budget explorer and the 5-question quiz.

Readings: IPCC AR6 Working Group I (2021), Summary for Policymakers: sections on carbon budgets; R for Data Science: data visualisation chapter